COM 3401
DATA COMMUNICATION AND NETWORKING

Lecture 06

TRANSMISSION MEDIA

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Topics covered

- Wired / Guided Media
  - Twisted Pair
  - Coaxial Cable
  - Optical Fiber
- Wireless / Unguided Media
  - Radio
  - Microwave
  - Infrared

Reference:

William Stallings, *Data and Computer Communications*, 8th Edition

Andrew S. Tanenbaum, *Computer Networks,* 5th Edition

6.1 Introduction



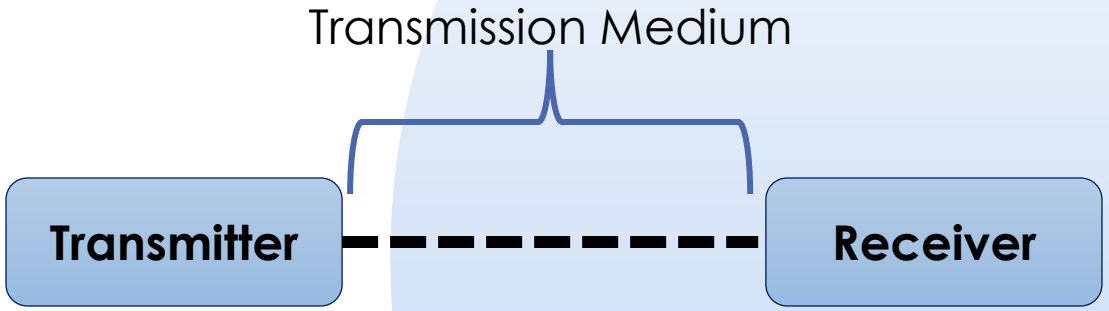
What is a Transmission Medium?

- A transmission medium is the physical path through which a signal travels from the transmitter to the receiver.

- Transmission media can be classified into:

1. Guided Media

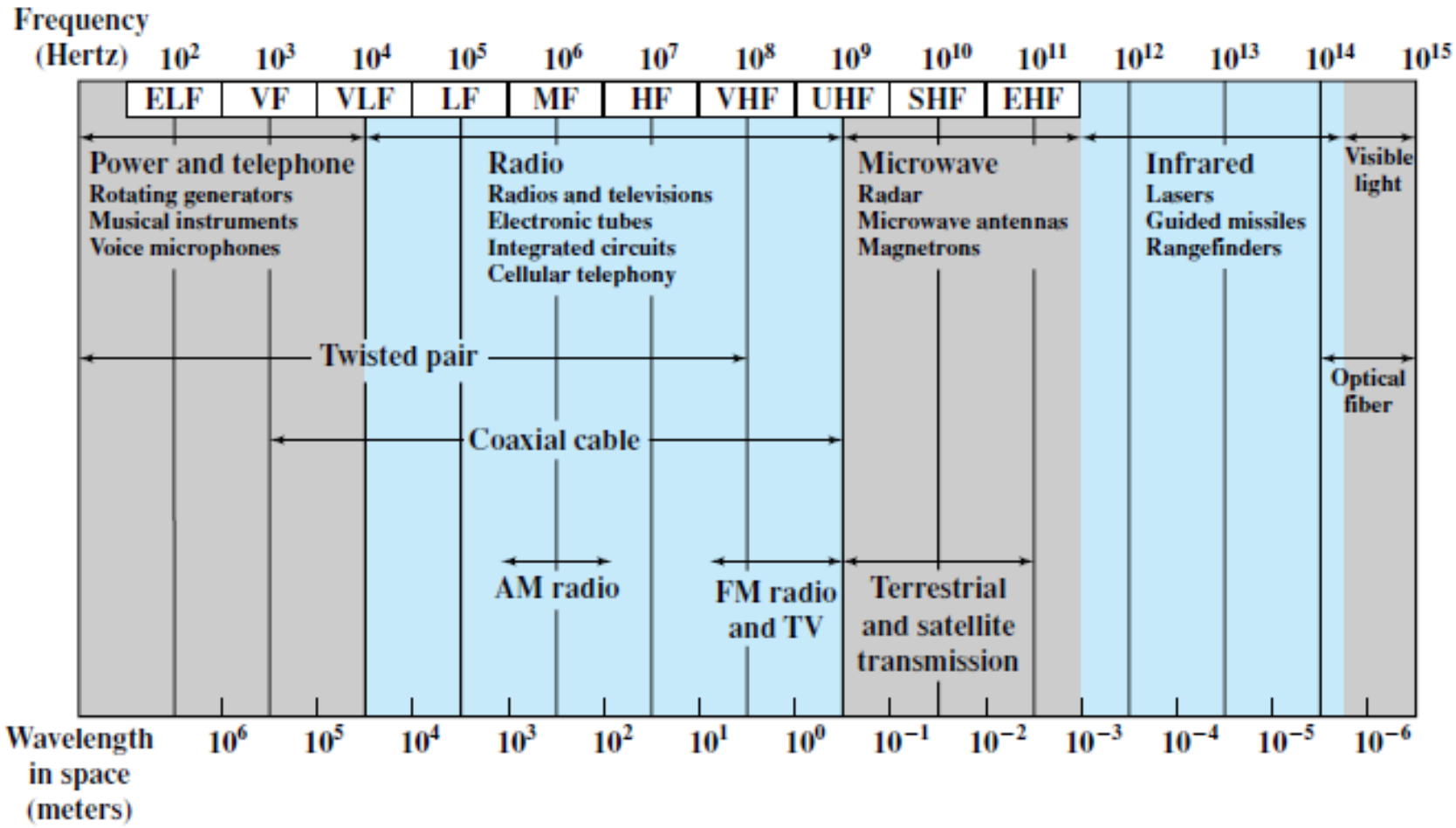
- Signals are guided along a physical path.
  - Examples:
    - Twisted pair
    - Coaxial cable
    - Optical fiber



2. Unguided / Wireless Media

- Signals propagate through air / vacuum/ other environments
  - Examples:
    - Radio
    - Microwave
    - Infrared

What is Transmission Medium?



ELF = Extremely low frequency
VF = Voice frequency
VLF = Very low frequency
LF = Low frequency
MF = Medium frequency
HF = High frequency
VHF = Very high frequency
UHF = Ultra high frequency
SHF = Super high frequency
EHF = Extremely high frequency

Factors Affecting the Choice of Medium

When choosing a transmission medium, important considerations include:

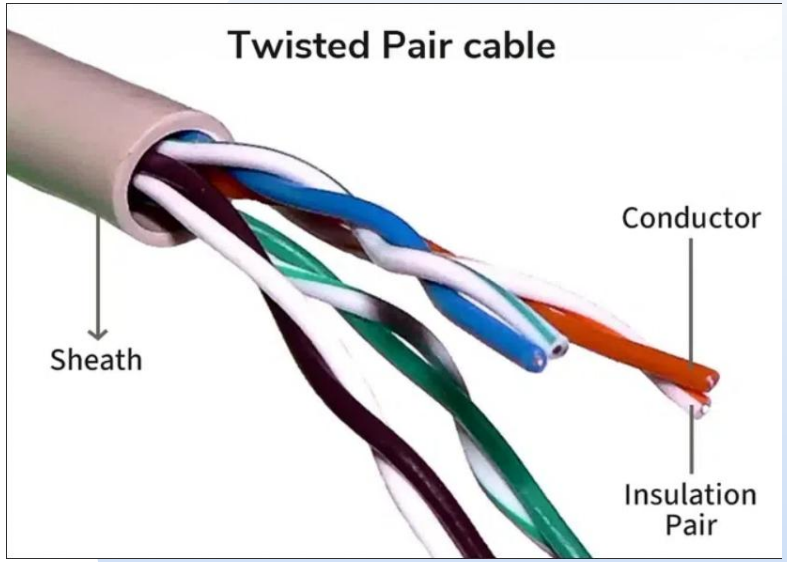
- **Bandwidth**
- **Data rate**
- **Distance**
- **Transmission impairments**
- **Interference**
- **Number of receivers**
- **Cost**
- **Application requirements**

6.2
Wired / Guided Media



Twisted Pair

- The least expensive and most widely used guided transmission medium
- A twisted pair consists of:
  - Two insulated copper wires
  - Wires are twisted together in a helical form.
  - A pair acts as a single communication link.
- Available in two types:
  1. STP (Shielded Twisted Pair)
  2. UTP (Unshielded Twisted Pair)



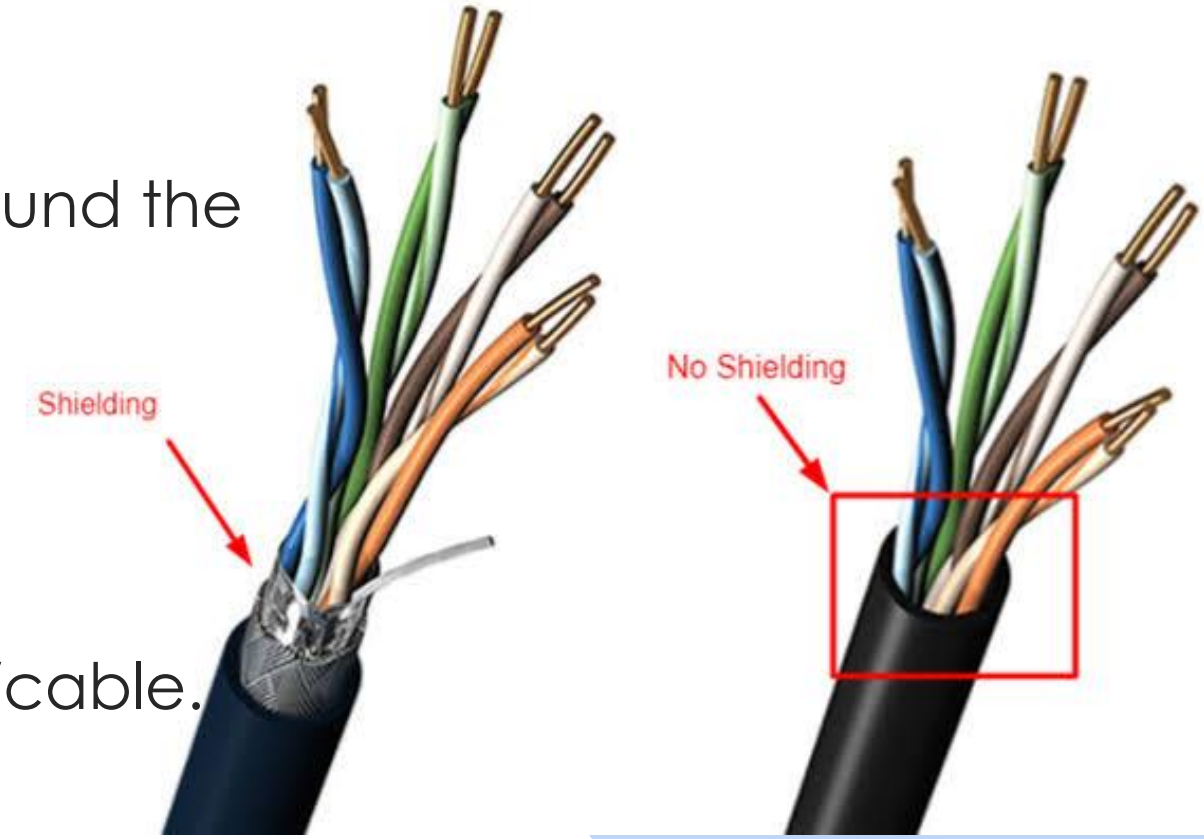
UTP and STP

Unshielded Twisted Pair — UTP

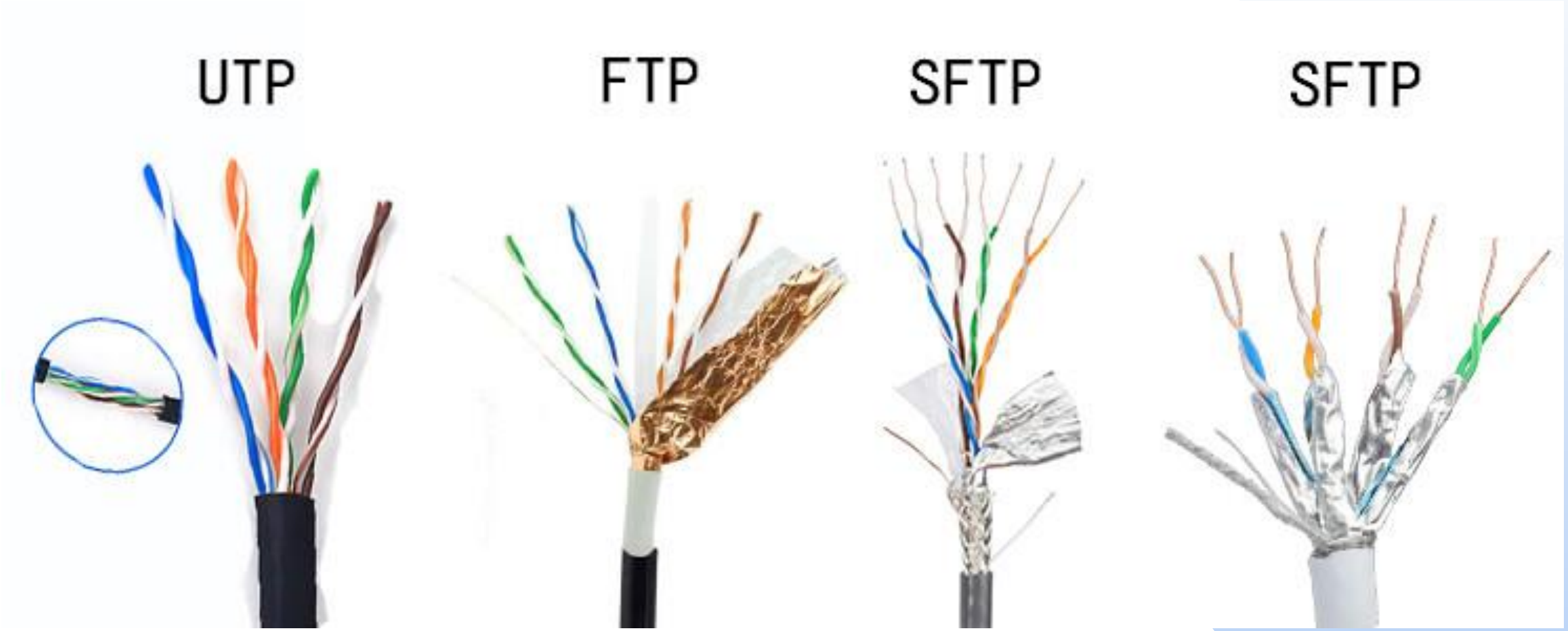
- Wires are twisted.
- No additional metallic shielding around the pairs.
- Common in buildings and LANs.

Shielded Twisted Pair — STP

- Includes shielding around the pairs/cable.
- Shielding reduces susceptibility to:
  - External interference
  - Crosstalk



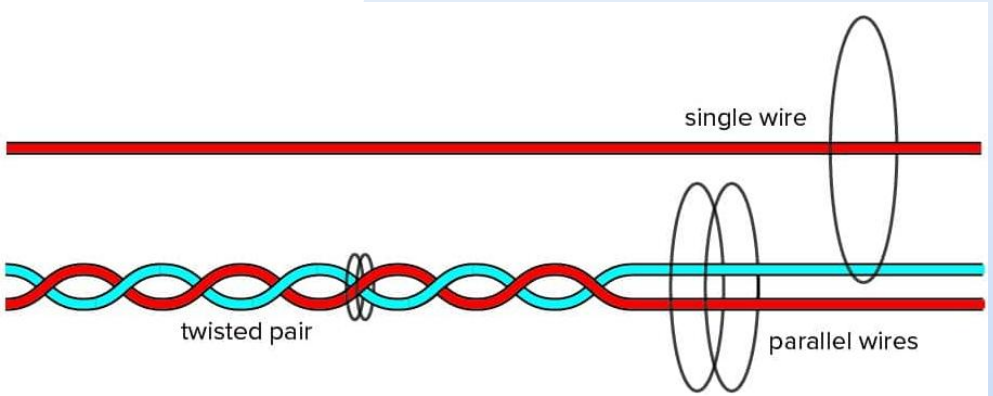
UTP and STP



Why are the Wires Twisted?

The problem

- Two parallel wires can behave like an antenna.
- They can:
  - Radiate signals
  - Pick up electromagnetic interference
  - Interfere with nearby pairs

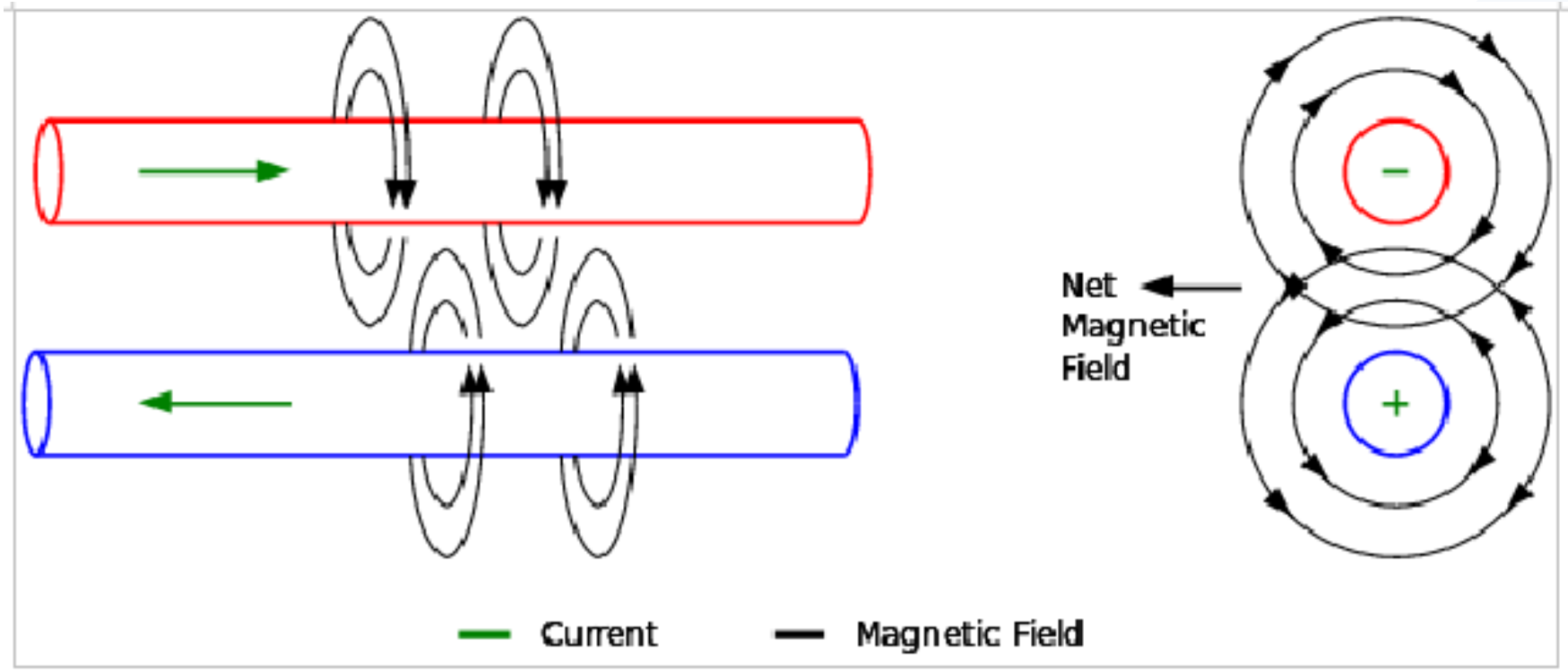


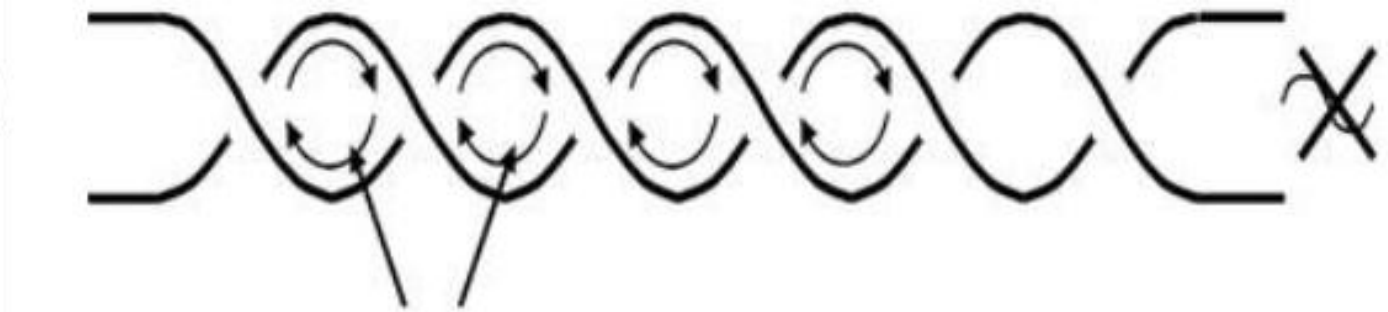
Twisting helps because:

- Waves from different twists tend to cancel.
- External noise tends to affect both wires similarly.
- The receiver uses the **difference in voltage** between the wires.

Therefore, twisting provides better immunity to external noise.

Why are the Wires Twisted?



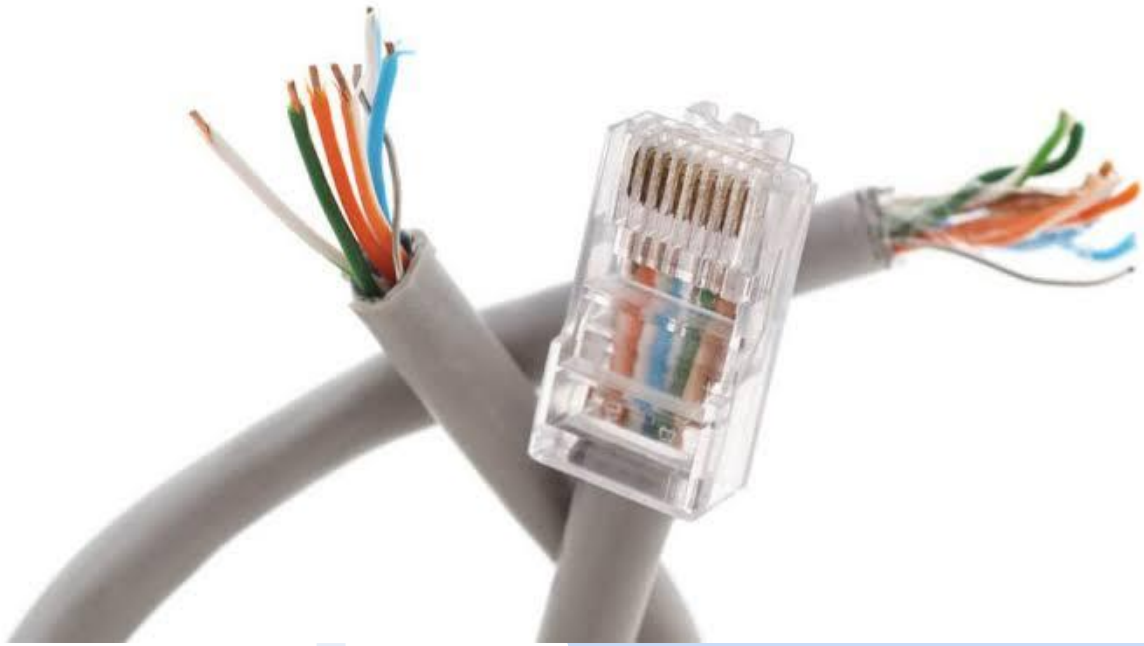


With twisted cable, the induced currents tend to be cancelled on the adjacent loops

Applications of Twisted Pair

Common Applications

- Telephone network
- Telephone subscriber loops
- ADSL
- Communications within buildings
- Local Area Networks
- Ethernet



- Telephone system as the most common application
- Twisted pair are the most common guided medium for both analog and digital signals.

Characteristics of Twisted Pair

Advantages

- Low cost
- Easy to install
- Widely available
- Can carry analog and digital information

Limitations

- Limited distance
- Limited bandwidth compared with coaxial cable and optical fiber
- More susceptible to interference and noise
- Crosstalk can be a concern

Twisted Pair: Performance

Characteristic	Twisted Pair
Frequency range	Up to about 1 MHz for multipair cable
Typical attenuation	Higher than coax/fiber
Repeater spacing	About 2 km
Cost	Low
Interference susceptibility	Relatively high

Actual performance depends on:

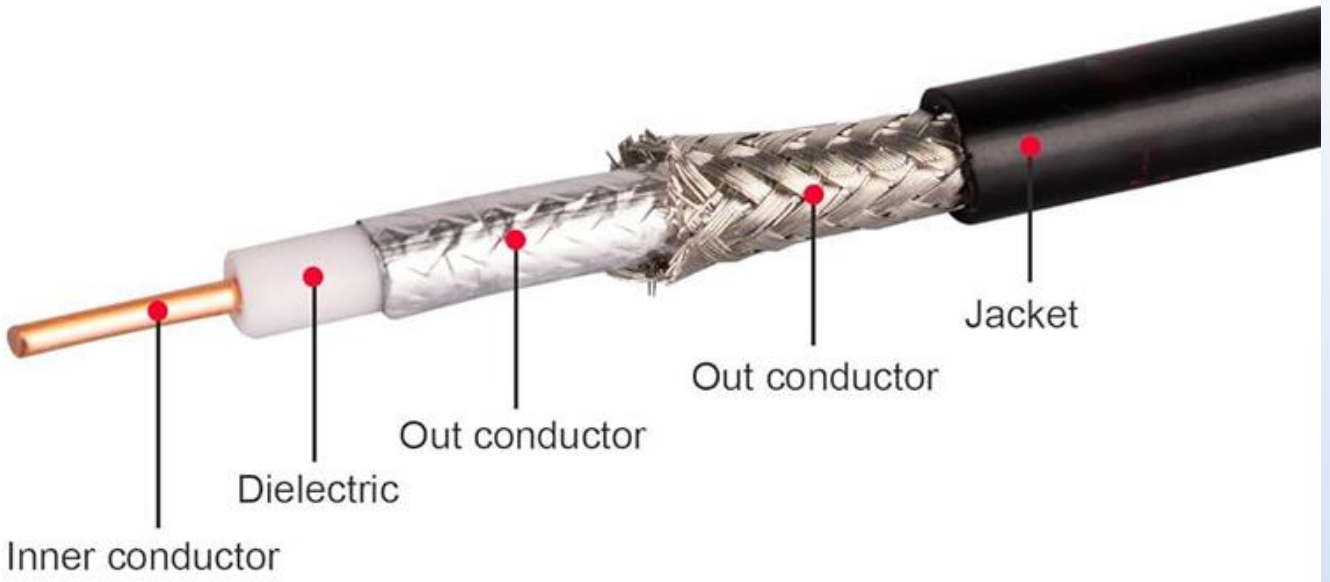
- Cable category
- Distance
- Signal type
- Installation

Coaxial Cable

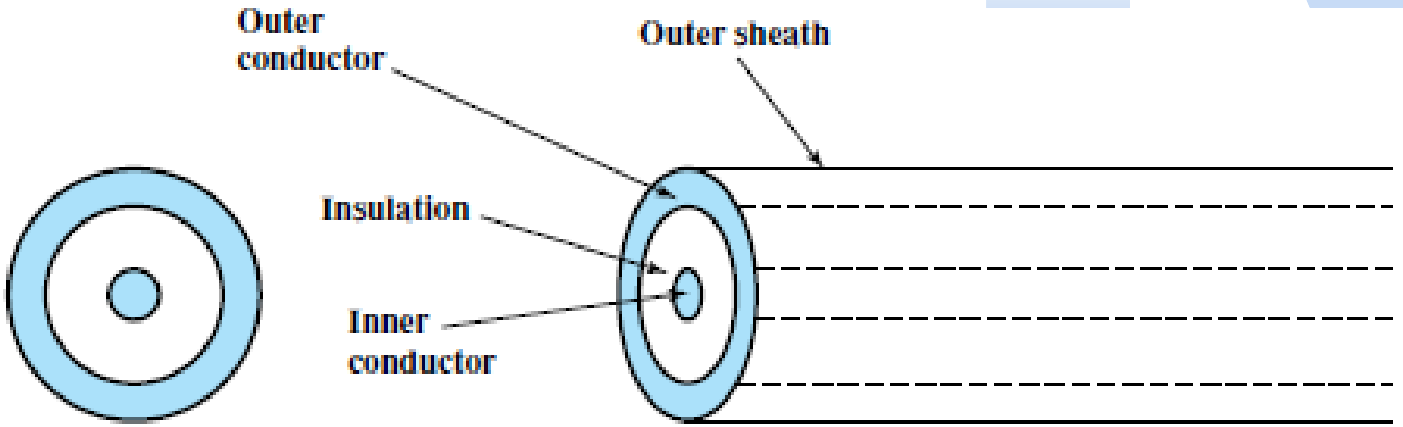
Physical Structure

Coaxial cable contains:

1. **Inner conductor**
2. **Insulating material**
3. **Outer conductor / shield**
4. **Protective outer jacket**



The outer conductor surrounds the inner conductor concentrically.



Why Coaxial Cable?

Compared with unshielded twisted pair, **Coaxial Cable provides:**

- Better shielding
- Greater bandwidth
- Longer-distance transmission
- Better noise immunity

- ❖ Coaxial cable is a versatile transmission medium, used in a wide variety of applications.
- ❖ Coaxial cable is used to transmit both analog and digital signals

Applications of Coaxial Cable

Major Applications

- Cable television
- Long-distance telephone transmission
- Short-run computer system links
- Local area networks

Historically:

- 50-ohm coaxial cable → digital transmission
- 75-ohm coaxial cable → analog transmission and cable television

Coaxial Cable: Characteristics

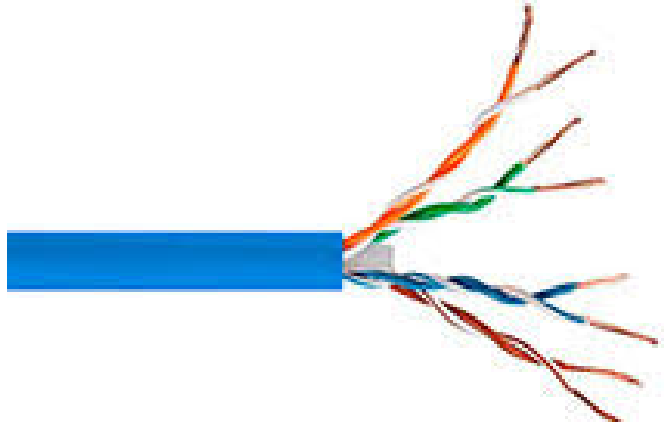
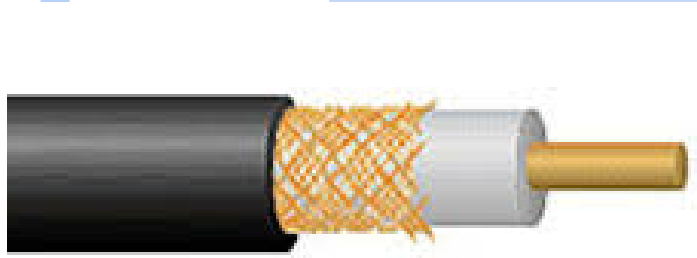
Advantages

- Higher bandwidth than UTP
- Better shielding
- Better noise immunity
- Longer distances
- Supports both analog and digital signals

Limitations

- More expensive than twisted pair
- Larger and less flexible
- Has attenuation
- Thermal noise can affect performance
- Intermodulation noise can occur when several channels use the cable

Coaxial vs Twisted Pair

Feature	Twisted Pair	Coaxial
Cost	Lower	Higher
Shielding	Less	Better
Bandwidth	Lower	Higher
Interference	More susceptible	Less susceptible
Distance	More limited	Greater
Construction	Two twisted wires 	Inner conductor + insulation + shield 

Optical Fiber

What is Optical Fiber?

- An optical fiber is a thin, flexible medium capable of guiding an optical signal.
- It is generally made using:
  - Glass
  - Plastic

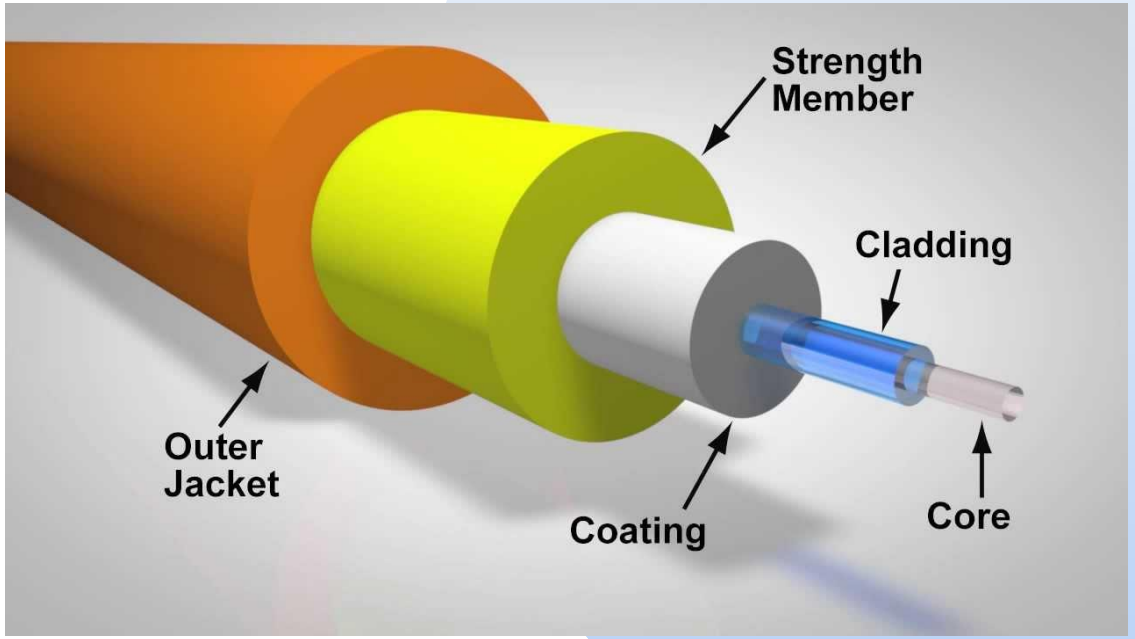


- A fiber-optic transmission system consists of:
  - **Light Source → Optical Fiber → Detector**

Structure of Optical Fiber

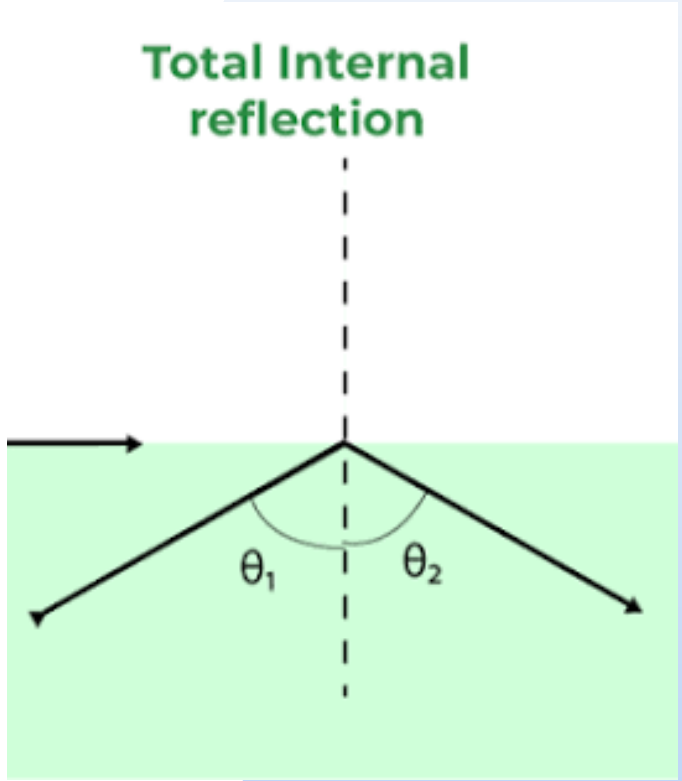
Three Main Parts

1. **Core**
   - Central region through which light travels.
2. **Cladding**
   - Surrounds the core.
   - Has different optical properties.
3. **Jacket**
   - Outer protective layer.
   - Protects against environmental damage.

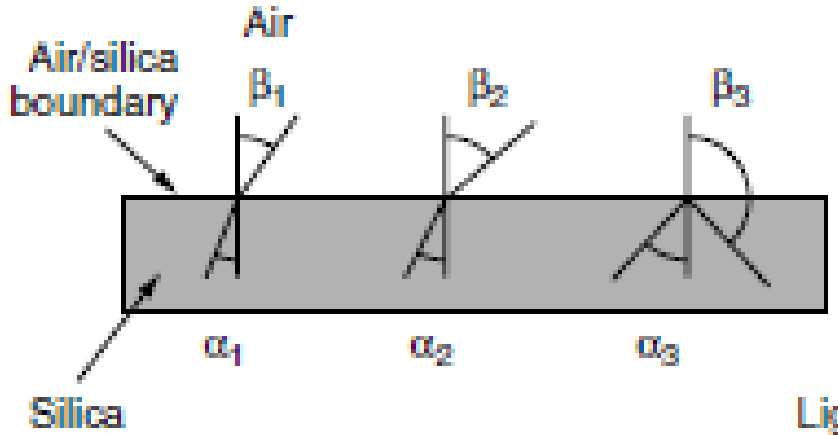


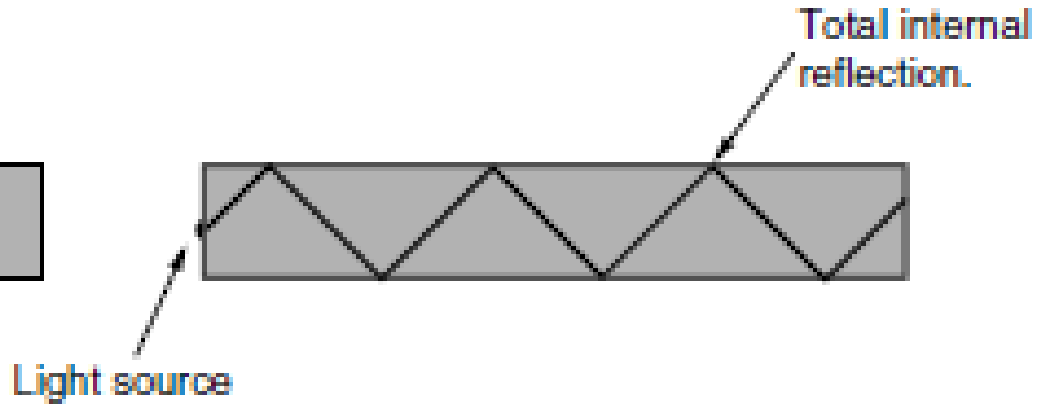
How Does Light Stay Inside the Fiber?

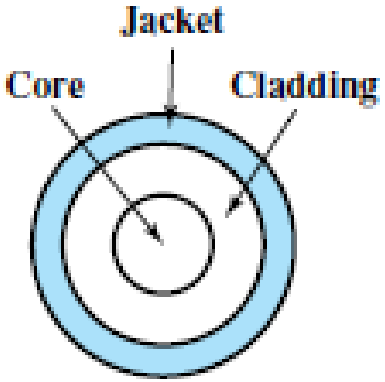
- When light reaches the boundary between two media:
  - It can be refracted.
- At sufficiently large angles of incidence, the light is reflected back into the fiber.
- This is called:
  - **Total Internal Reflection**
- The light can therefore remain trapped inside the fiber and travel long distances.

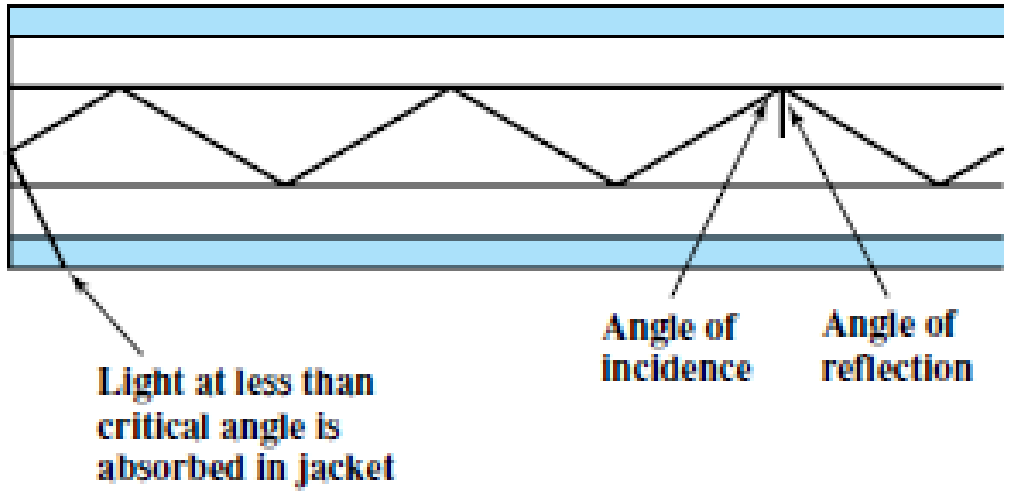


How Does Light Stay Inside the Fiber?





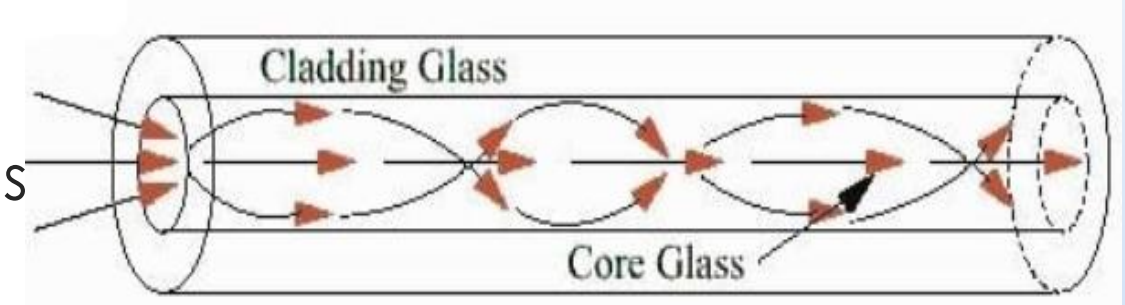




Multimode and Single-Mode Fiber

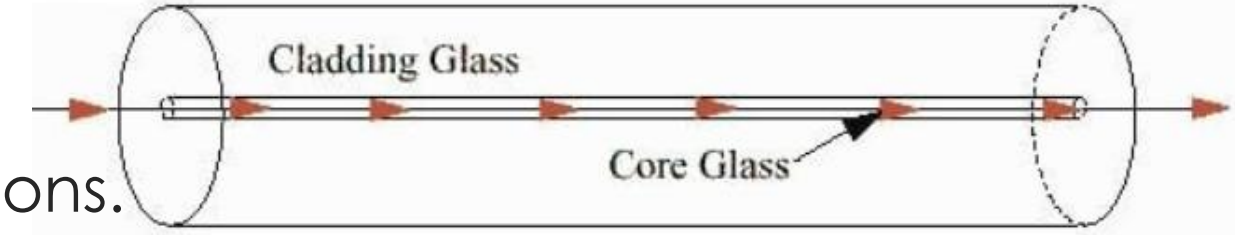
Multimode Fiber

- Multiple rays/modes can propagate through the fiber.
- Different rays follow different paths.
- Used for shorter-distance applications



Single-Mode Fiber

- Fiber diameter is reduced to only a few wavelengths.
- Light propagates essentially along a single path.
- More expensive.
- Used for longer-distance applications.

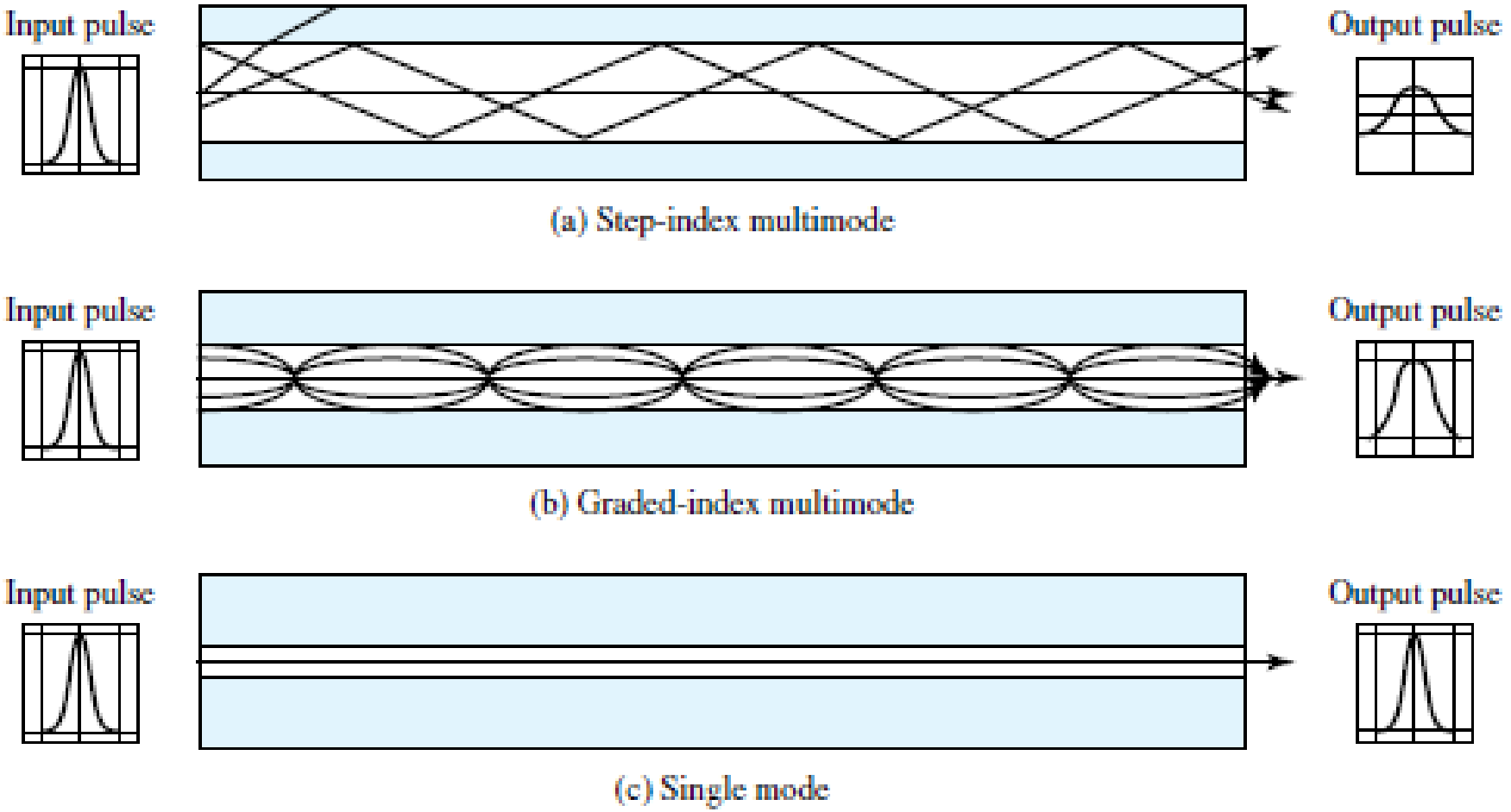


Types of propagation in fiber optic cables

There are three types:

1. Step-index multimode:
   - Rays at shallow angles are reflected and propagated along the fiber
   - Other rays are absorbed by the surrounding material
   - Due to multiple paths signal elements are spread out in time
2. Graded-index multimode:
   - Obtained by varying the index of refraction of the core
   - The higher refractive index at the center; rays advances more slowly
   - Rather than zig-zagging off the cladding, light in the core curves helically because of the graded index
3. Single mode:
   - Obtained by reducing the radius of the core to the order of a wavelength, only a single ray (axial ray) can pass

Types of propagation in fiber optic cables



Light Sources and Detector

Transmitters

- Two common light sources:

1. LED	2. Semiconductor Laser
Lower cost	More expensive
Lower data rate	Higher data rate
Longer life	Shorter life
Lower temperature sensitivity	Higher temperature sensitivity

Receiver

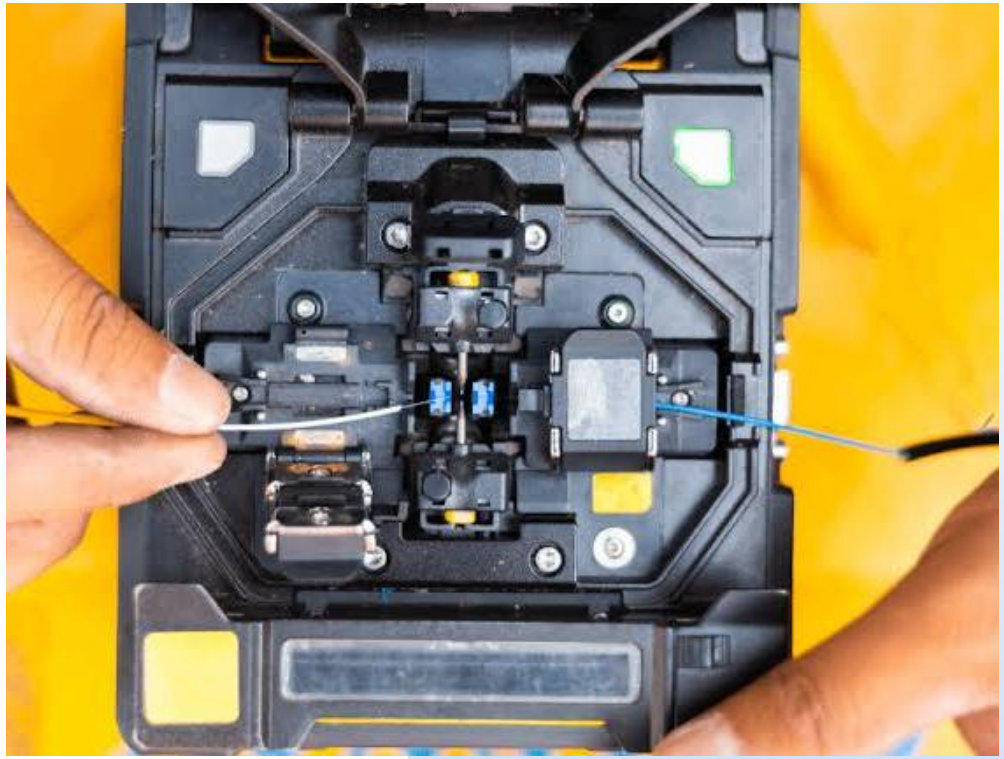
- Uses a **photodiode**
- Converts optical pulses back into electrical signals.

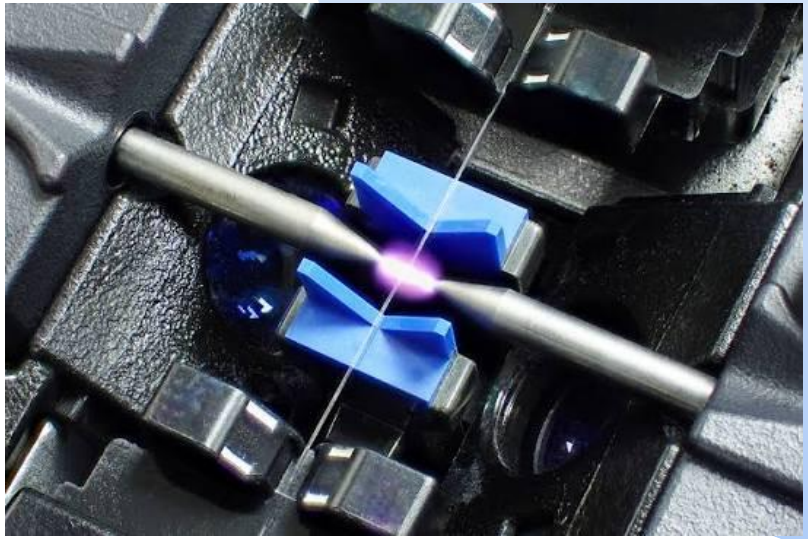
Advantages of Optical Fiber

- Very high bandwidth
- Low attenuation
- Long repeater spacing
- Small size
- Low weight
- Immune to electromagnetic interference
- Not affected by crosstalk
- Difficult to tap
- High security against eavesdropping

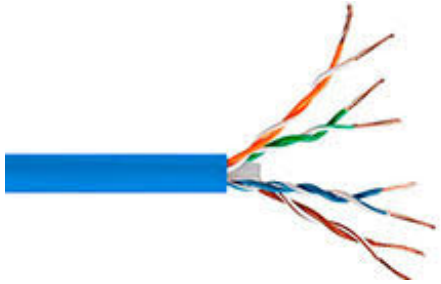
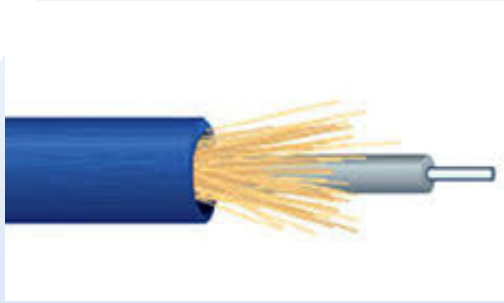
Limitations of Optical Fiber

- Requires specialized skills
- Can be damaged by excessive bending
- Optical transmission is inherently unidirectional
- Two-way communication requires:
  - Two fibers, or
  - Two frequency bands on one fiber





Comparing the Three Wired Media

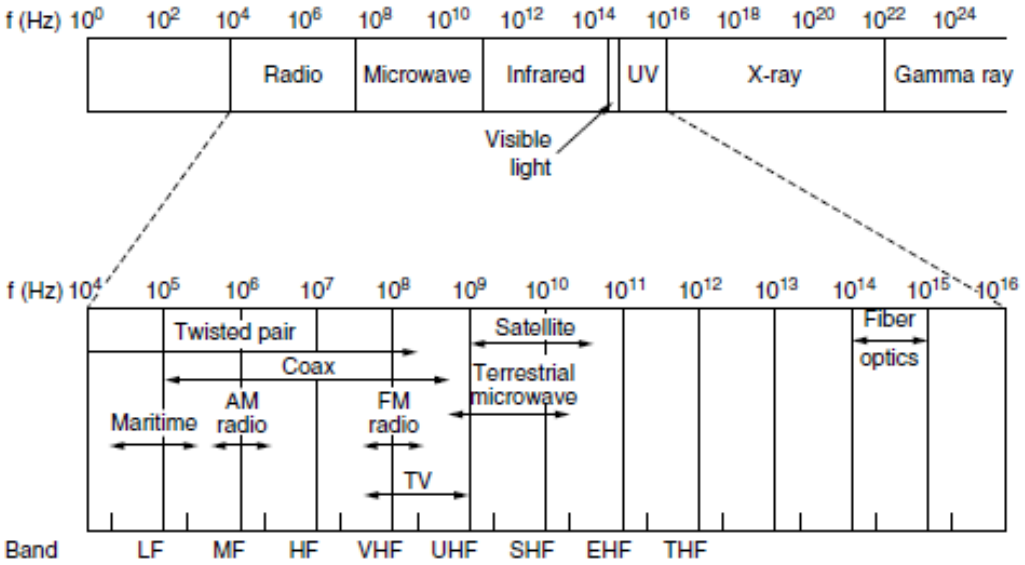
	Twisted Pair	Coaxial	Optical Fiber
			
Medium	Copper	Copper	Glass/plastic
Bandwidth	Lowest	Higher	Very high
Interference	Higher	Lower	Very low
Attenuation	Higher	Moderate	Low
Cost	Low	Moderate	Higher/specialized
Distance	Limited	Greater	Very long
Typical use	Telephone/LAN	Cable TV/legacy links	Backbone/high-speed links



6.3
Wireless / unguided Media

Wireless Transmission

- Signals propagate without a physical conductor.
- Common techniques include:
  - Radio
  - Microwave
  - Infrared
- Wireless transmission depends strongly on:
  - Frequency
  - Bandwidth
  - Directionality
  - Propagation characteristics



Radio Transmission

Radio frequency waves:

- Are relatively easy to generate.
- Can travel long distances.
- Can penetrate buildings.
- Are widely used indoors and outdoors.
- Are generally **omnidirectional**.



Therefore:

- Transmitter and receiver do not necessarily need precise physical alignment.

Radio: Frequency and Propagation

Radio-wave behavior depends on frequency.

Low Frequencies

- Better obstacle penetration
- Signal power decreases with distance
- Lower bandwidth

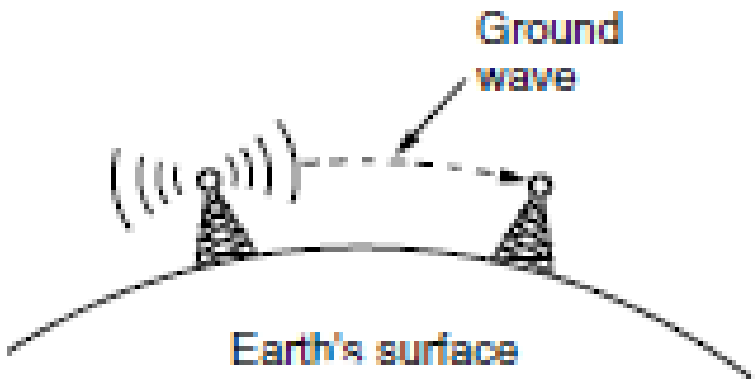
Higher Frequencies

- Tend to travel more nearly in straight lines
- Can reflect from obstacles
- More affected by rain and other obstacles

Radio: Propagation Modes

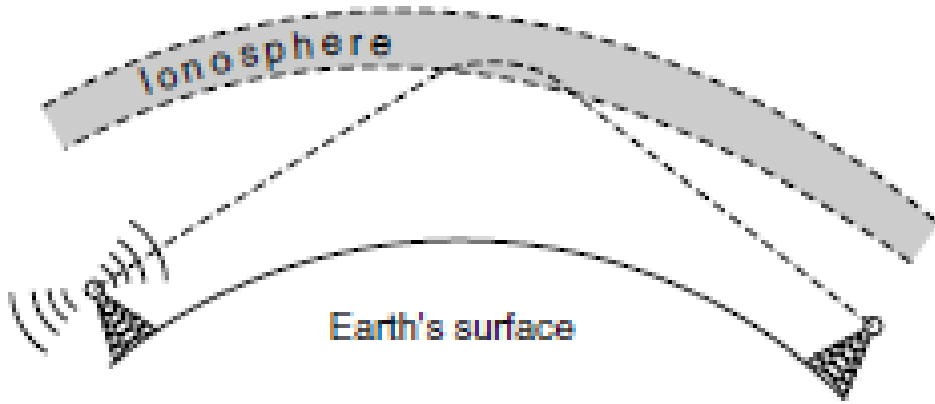
1. Ground Wave

- Used mainly at lower frequencies (VLF, LF, MF bands).
- Follows the curvature of the Earth.
- AM radio is an example.



2. Sky Wave

- Used in HF/VHF ranges.
- Waves can be refracted by the ionosphere.
- Can support long-distance communication.



Radio: Advantages and Limitations

Advantages

- Easy to generate
- Long-distance capability
- Can penetrate buildings
- Omnidirectional

Limitations

- Interference
- Path loss
- Multipath effects
- Frequency-dependent behavior
- Limited bandwidth at lower frequencies

Microwave Transmission

Microwave signals:

- Travel approximately in straight lines.
- Can be focused into narrow beams.
- Use directional antennas.
- Require accurate transmitter/receiver alignment.
- A common antenna is the **Parabolic Antenna / Dish**
- Two types:
  1. Terrestrial microwave
  2. Satellite microwave

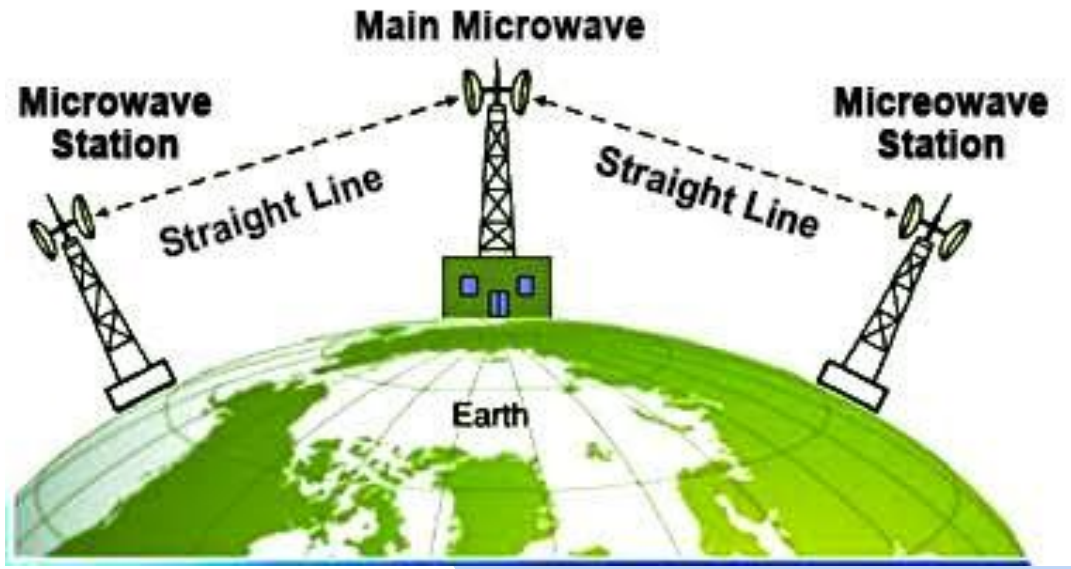


Terrestrial Microwave

How it works

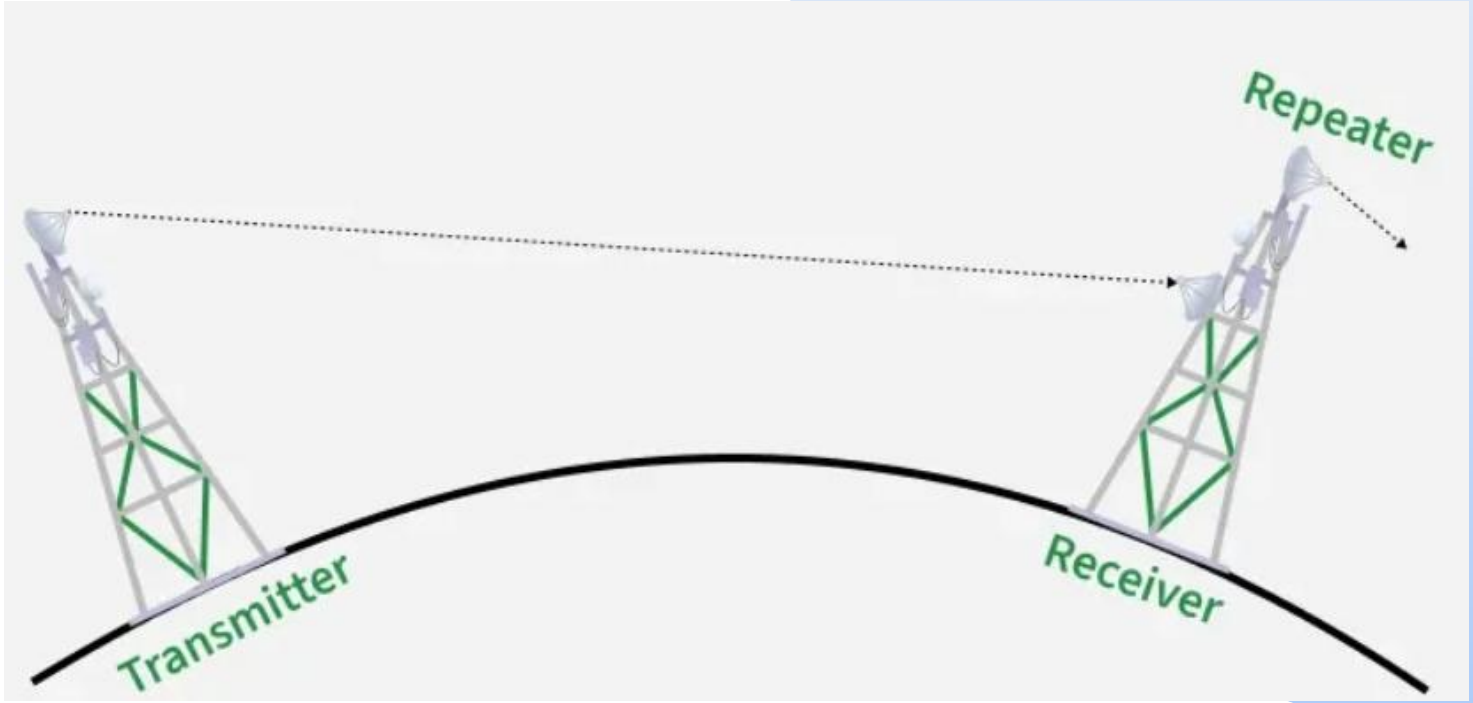
Microwave tower ➡️ Directional beam ➡️ Microwave tower ➡️ Relay ➡️ Next tower

- Microwave towers are positioned high enough to achieve **line-of-sight transmission**.
- Repeaters are needed periodically because the Earth eventually obstructs the straight-line path.
- Typical point-to-point terrestrial microwave systems using relay towers and line-of-sight transmission.



Terrestrial Microwave: Applications

- Long-distance telecommunications
- Voice transmission
- Television transmission
- Point-to-point links between buildings
- LAN interconnection
- Cellular systems



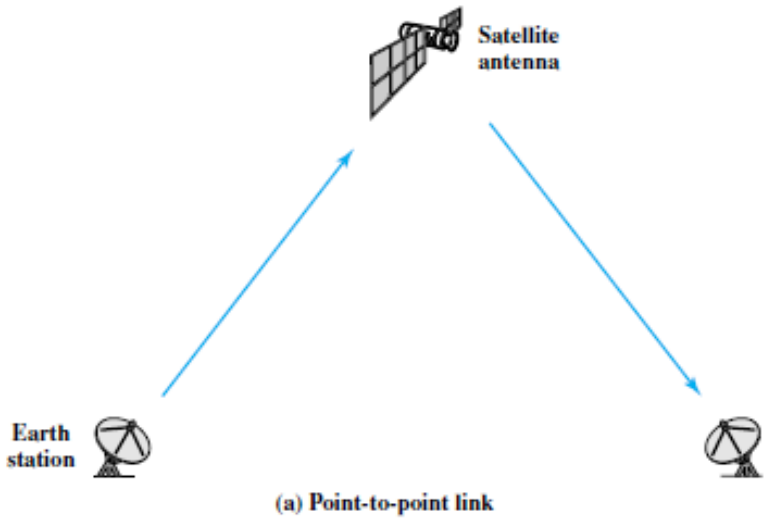
Satellite Microwave

- A communication satellite is used as a microwave relay station.
- It is used to link two or more ground-based microwave transmitter/receivers, known as earth stations, or ground stations.
- The satellite receives transmissions on one frequency band **(uplink)**, amplifies or repeats the signal, and transmits it on another frequency **(downlink)**.
- A single orbiting satellite will operate on a number of frequency bands, called **transponder channels**, or simply **transponders.**
- For a communication satellite to function effectively, it is generally required that it remain stationary with respect to its position over the earth.
- Two common configurations:
  1. Point to point link
  2. Broadcast link

Satellite Microwave configurations

1. Point-to-Point Link

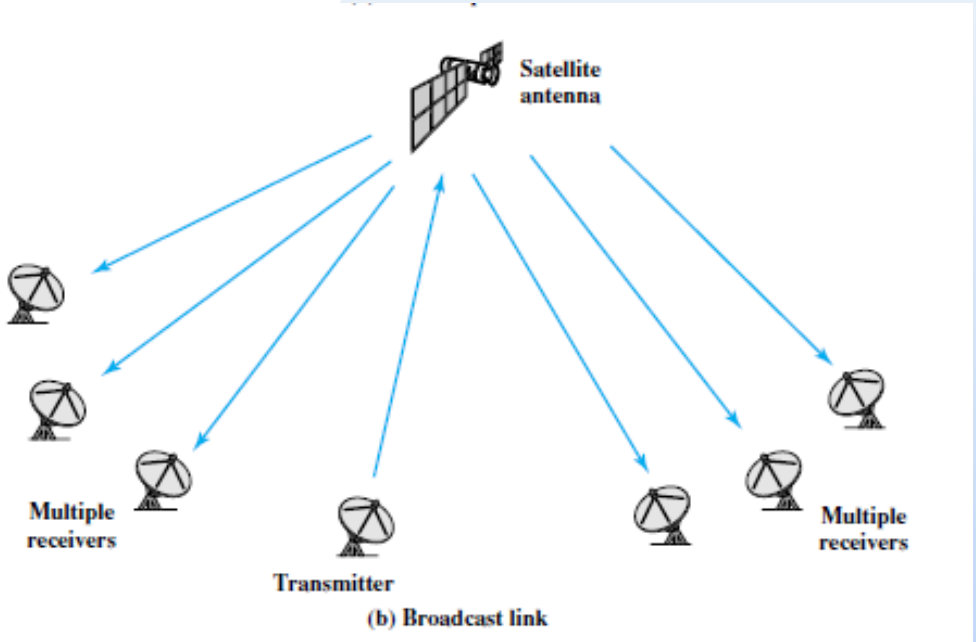
- A **point-to-point satellite link** provides communication between **one specific transmitting station and one specific receiving station** through a satellite.
- **Example:**
  - **Station A → Satellite → Station B**
- One transmitter
- One intended receiver
- Communication is between two specific locations.
- The satellite provides the communication path between them.



Satellite Microwave configurations

2. Broadcast Link

- A **broadcast satellite link** allows one transmitting station to send information to **many receiving stations simultaneously**.
- **Example:**
  - **Station A → Satellite → Multiple Receivers**
- **Key idea:**
  - **One transmitter → Many receivers**
- The same signal can be received by multiple users within the satellite's coverage area.



Satellite Microwave: Applications

- Television distribution
- Long-distance telephone transmission
- Private business networks
- Global positioning

Microwave: Advantages and Limitations

Advantages

- High potential bandwidth
- High potential data rates
- Directional beams
- Fewer repeaters than comparable long cable links
- Useful where cables are difficult to install

Limitations

- Requires line of sight
- Accurate antenna alignment
- Buildings can obstruct transmission
- Multipath fading
- Rain attenuation
- Interference
- Frequency allocation is regulated

Infrared Transmission

- Infrared communication uses electromagnetic waves closer to visible light.
- **Characteristics**
  - Short-range communication
  - Relatively directional
  - Cheap
  - Easy to build
  - Does not pass through solid objects



Television



Infrared Radiations



Remote

Infrared: Applications

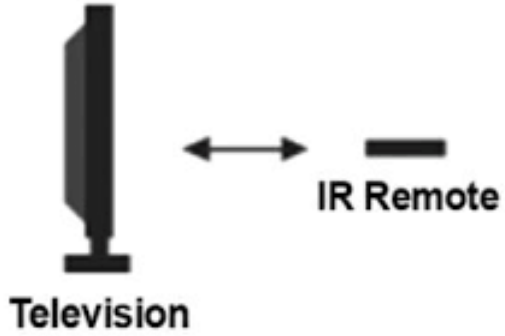
- Television remote controls
- VCR/stereo remote controls
- Short-range desktop communication
- Notebook-to-printer communication
- IrDA

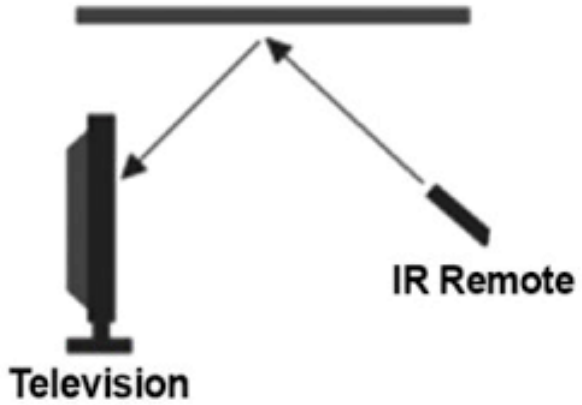


Infrared: Advantages and Limitations

Advantages

- Short-range communication
- Relatively directional
- Cheap
- Easy to build
- Does not interfere easily with adjacent rooms
- Better resistance to eavesdropping than radio
- No government license required





Limitation

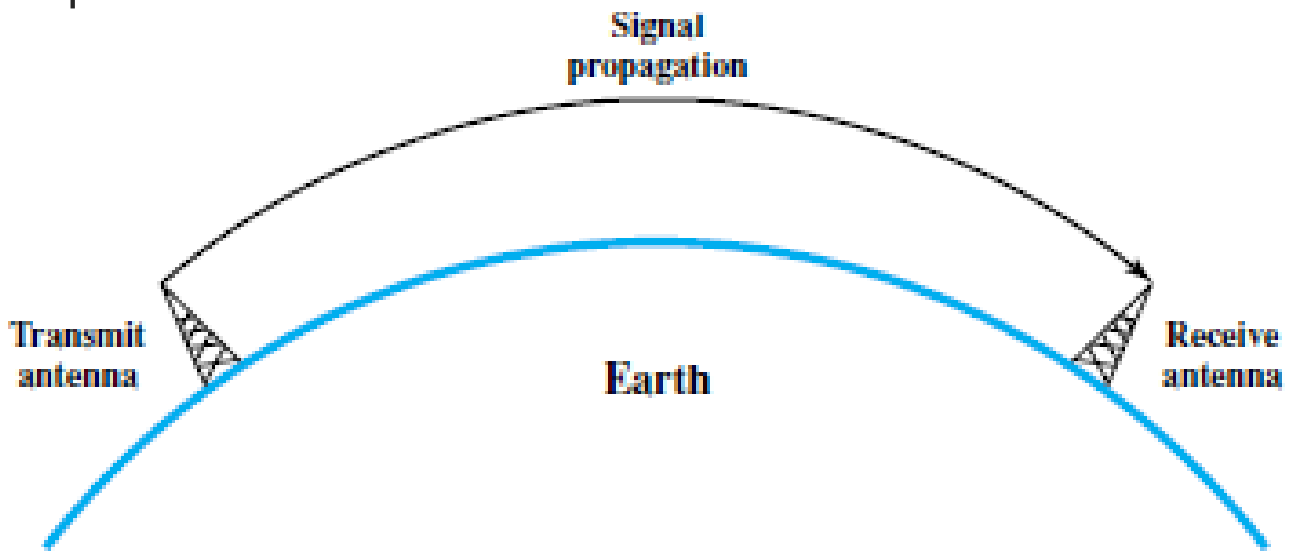
- Cannot pass through solid walls/objects.

Wireless Propagation

A signal radiated from an antenna travels along one of three routes:

1. Ground wave

- Follows the contour of earth
- Can propagate considerable distance over visual horizon
- Found in frequencies up to 2MHz

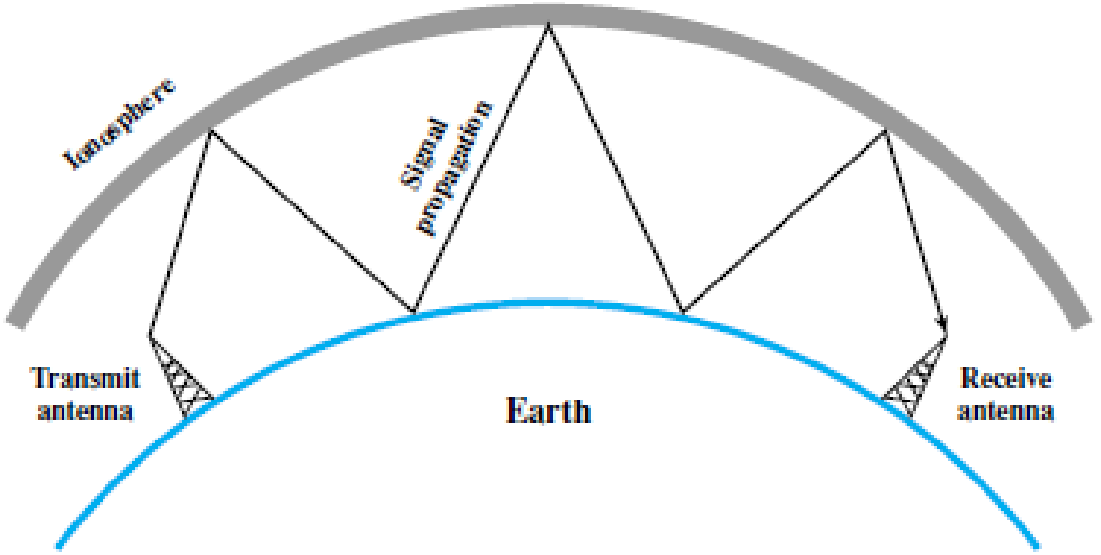


(a) Ground wave propagation (below 2 MHz)

Wireless Propagation

2. Sky wave

- a signal from an earth-based antenna is reflected from the ionized layer of the upper atmosphere (ionosphere) back down to earth
- Can travel through no of hops enabling it to be picked up thousand of kilometers away
- Found in frequencies 2MHz to 30 MHz.

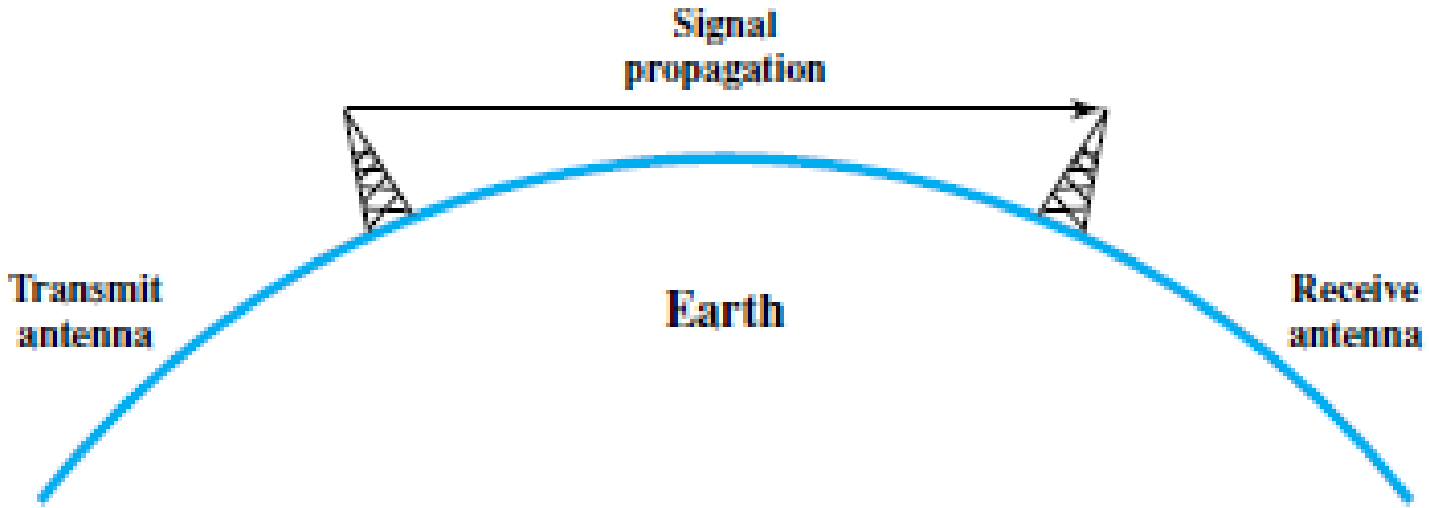


(b) Sky wave propagation (2 to 30 MHz)

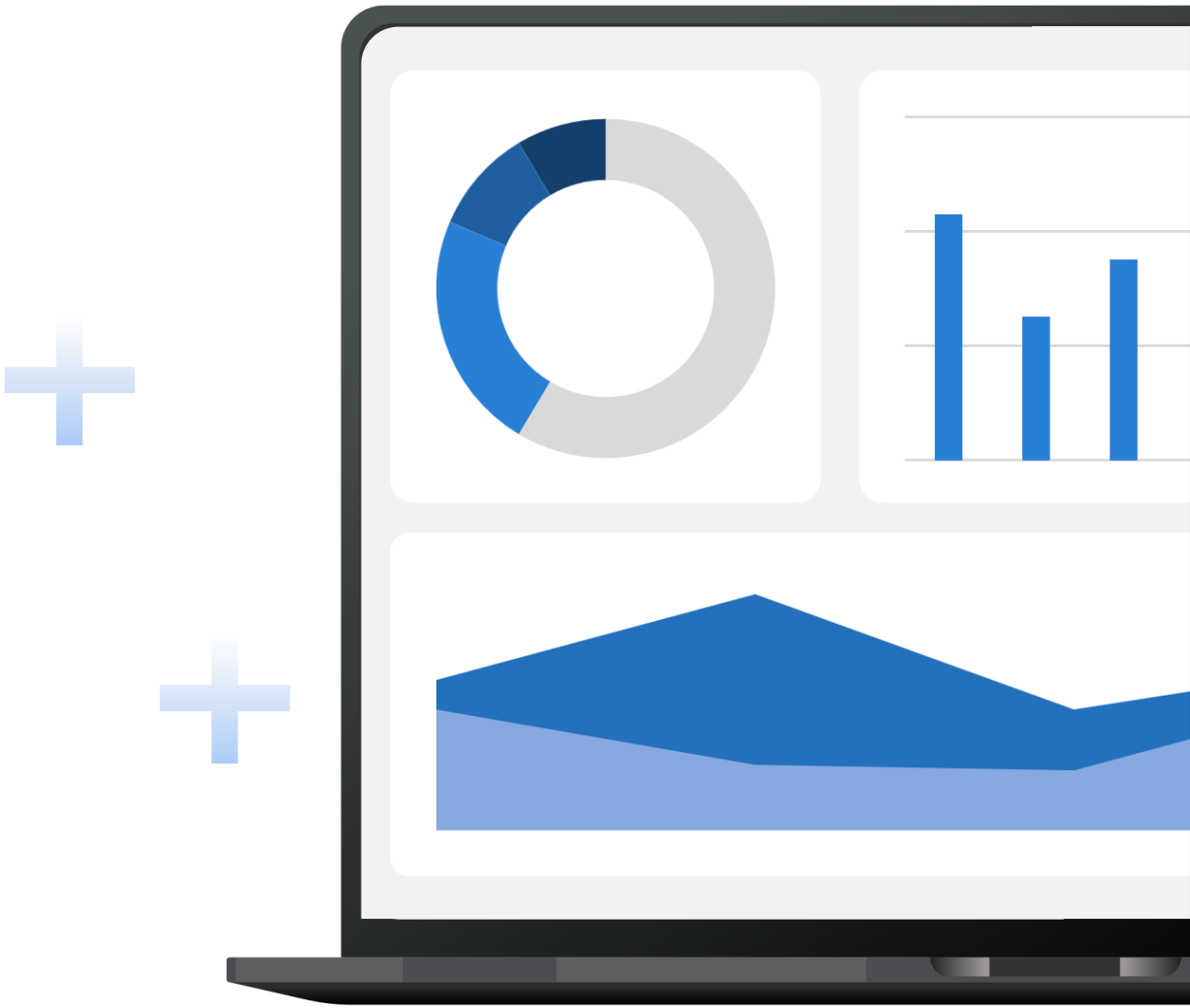
Wireless Propagation

3. Line of sight (LOS)

- Above 30 MHz, neither ground wave nor sky wave propagation modes operate, and communication must be by line of sight
- transmitting and receiving antennas must be within an *effective* line of sight of each other



(c) Line-of-sight (LOS) propagation (above 30 MHz)



6.4 Comparisons

Guided vs Wireless Media

Aspect	Guided Media	Wireless Media
Transmission	Signal travels through a **physical path**	Signal travels through **free space**
Examples	Twisted Pair, Coaxial, Optical Fiber	Radio, Microwave, Infrared
Direction	Defined by the physical cable	Omnidirectional or directional
Bandwidth	Depends on medium; **fiber offers very high bandwidth**	Depends on frequency and propagation conditions
Interference	Copper is susceptible; **fiber is highly resistant**	More exposed to interference and propagation effects
Security	Physical access generally required to intercept	Easier to intercept because signals propagate through open space
Mobility	Limited — devices depend on physical connections	High — supports mobile/portable devices
Installation	Requires physical cabling and infrastructure	Easier where cabling is difficult or impractical
Typical Use	LANs, telephone networks, cable TV, backbone links	Radio communication, microwave links, remote/mobile communication

Wired vs Wireless

	Wired / Guided	Wireless / Unguided
Physical path	Yes	No
Examples	TP, coax, fiber	Radio, microwave, infrared
Mobility	Limited	High
Interference	Depends on medium	Important concern
Installation	Requires cabling	No physical cable
Directionality	Determined by physical path	Depends strongly on frequency/antenna

Transmission Media: Overall Comparison

Medium	Main Strength	Main Limitation	Typical Application
Twisted Pair	Low cost	Limited bandwidth/distance	Telephone, LAN
Coaxial	Better bandwidth/shielding	Larger, more costly	Cable TV
Optical Fiber	Very high capacity	Specialized installation	Long-haul/backbone
Radio	Long-distance/penetrates buildings	Interference	Radio/wireless
Microwave	High bandwidth/directional	Line of sight	Point-to-point links
Infrared	Cheap, short-range, localized	Cannot pass through solids	Remote controls/short-range

How Do We Choose a Transmission Medium?

If you were designing a network, which medium would you choose?

Consider:

- Required data rate
- Required distance
- Cost
- Installation requirements
- Interference
- Mobility
- Security
- Environment
- Reliability

Scenarios:

- Office LAN → ?
- Long-distance backbone → ?
- Remote control device→ ?
- Building-to-building link → ?
- Telephone subscriber loop → ?

Review Questions

1. What is a transmission medium?
2. What is the difference between guided and unguided media?
3. Why are wires twisted in twisted-pair cable?
4. What is the difference between UTP and STP?
5. Why does coaxial cable provide better noise immunity than twisted pair?
6. What are the three main parts of an optical fiber?
7. What is total internal reflection?
8. Distinguish between multimode and single-mode fiber.
9. Why are microwave antennas directional?
10. What is line-of-sight transmission?
11. What is the main difference between radio and microwave?
12. Why is the inability of infrared to pass through walls both an advantage and a disadvantage?

Thank you

COM 3401 – DATA COMMUNICATION AND NETWORKING

Lecture 06

TRANSMISSION MEDIA